

P S Mithul Sourav

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Education

National Institute of Technology Karnataka (NITK), Surathkal
B.Tech. in Mechanical Engineering; CGPA: 7.37/10

Aug 2023 – Jun 2027
Surathkal, India

Research Interests

I am interested in building robots and mechatronic systems and in understanding the dynamics and control that govern how they move and interact with the real world. I am particularly interested in experimental and field robotics, where mechanical design, sensing, vibration, and control come together in physical hardware.

Research Projects

Variable-Stiffness Suspension and Vibration-Aware Control for a Rough-Terrain Rover

Sep 2026 – Present

- Designed a six-wheel rough-terrain rover to investigate whether **adaptive suspension stiffness** can reduce vibration-induced speed limitations on rough terrain.
- Developed **soft, medium, and stiff suspension states** with mechanical locking and a separate load path so the rover remains mechanically supported if the stiffness-changing mechanism fails.
- Defined an **IMU- and encoder-based control strategy** to detect sustained vibration, change stiffness, verify the new state, and reduce speed only when reconfiguration is insufficient; defined a validation plan comparing fixed-stiffness, speed-only, stiffness-only, and combined control.

Modular Terrain-Adaptive Robot for Leaf Collection on Wet and Undulating Terrain

Jul 2026 – Present

- Framed the field-robotics problem around collecting **dry and wet leaves from uneven terrain** while minimizing unwanted soil pickup.
- Decomposed the platform into **locomotion, pickup, transfer, storage, and terrain-following** subsystems; screened pickup concepts against space, manufacturability, and integration constraints and developed the bill of materials (BOM) around the most practical options.
- Defined performance tests for **traction, pickup efficiency, soil rejection, and endurance** to compare design iterations using measured system performance rather than subjective preference.

Tensegrity-Based Variable-Stiffness Knee Joint for a Knee Exoskeleton

May 2026 – Present

- Modeled a tensegrity-based variable-stiffness knee joint for a **knee exoskeleton**, studying how geometry and internal loading alter joint stiffness.
- Analyzed **force output and joint stiffness in MATLAB** while varying geometry, external loading, and internal force.
- Identified the design variables with the strongest influence on force and stiffness and narrowed the design space to a smaller set of configurations for a **physical prototype of the knee exoskeleton**.

Additional Projects

Physics-Informed Multi-Fidelity FEM Surrogate for Reaction-Wheel-Induced Spacecraft Microvibration and Payload Jitter

Aug 2026 – Oct 2026

- Developed a **parameterized high-fidelity finite-element model** of the reaction-wheel-mount-spacecraft-payload system with multi-order speed-dependent excitation, modal, harmonic-response, and rotordynamic analyses, and payload line-of-sight jitter outputs; identified a strong response near **5,800 rpm** with only **0.33%** mesh sensitivity in peak amplitude.
- Verified the model through **analytical, mesh, frequency-step, modal-truncation, and dynamic-equilibrium checks**; built low-fidelity and reduced-order models and a leakage-safe multi-fidelity dataset, then compared **MLP, DeepONet, and physics-/multi-fidelity variants** on resonance accuracy, physics consistency, OOD generalization, and data efficiency before using the surrogate for uncertainty quantification, resonance-risk mapping, and robust design.

Actuated Knee-Assistance System

Oct 2025 – Sep 2026

- Developed a passive spring-assisted knee concept and refined the mechanism in **SolidWorks** and **ANSYS**, using interference checks to identify layouts capable of moving through the intended flexion range.
- Extended the concept into an **actuated cable-and-spring brace** with knee-angle tracking and defined the drive, sensing, power, manual-release, and bench-test architecture; the team placed in the **top 5 of 70 teams nationwide** in the Incubate X Prosthetic Challenge.

Uncertainty-Aware Navigation for an Indoor Mobile Robot

Sep 2025 – Dec 2025

- Implemented a differential-drive simulator with **A* path planning, heading control, wheel encoders, an IMU, and EKF localization**, then injected encoder noise, IMU drift, yaw bias, and wheel slip.
- Added **confidence-aware speed and recovery logic** and evaluated the navigation stack through Monte Carlo trials using collisions, localization error, recovery events, success rate, and time to goal.

Selected Work Experience

Vayu Aerospace Pvt. Ltd.

Jun 2026 – Jul 2026

Product Intern – UAV Vibration Analysis & Isolation

Bengaluru, India

- Investigated vibration transmission from the motors through the airframe to the flight-controller IMU; modeled rigid, elastomer-isolated, and suspended mounts in **ANSYS** and compared natural frequencies, mode shapes, and vibration response.
- Integrated and ground-tested the shortlisted mounts, then analyzed IMU time histories, **RMS** levels, and **FFT** spectra in **MATLAB**; combined simulation, integration, and test evidence to support selection of the elastomer-isolated tray.

Thinkify Labs

May 2025 – Aug 2025

Product and Strategy Intern

Bengaluru, India

- Developed an **AI-based lead-qualification workflow** that converted profile and email information into a repeatable screening process.
- Defined routing logic so qualified prospects reached the appropriate sales representatives, then refined the workflow into a structured, testable process that contributed to a reported **25% improvement in operational efficiency**.

Technical Skills

CAD & CAE: SolidWorks; PTC Creo; ANSYS Mechanical (modal and harmonic analysis); COMSOL Multiphysics.

Programming & Computation: MATLAB; Python; C; C++.

Leadership & Recognition

Indian Society for Technical Education (ISTE), NITK – Secretary

Sep 2024 – Present

- Led sponsor outreach, budgeting, vendor coordination, and logistics for a technical event with 500+ participants; secured three sponsors and coordinated execution across teams.

Global Case Competition at Harvard – Global Top 50

Feb 2026 – Mar 2026

- Analyzed the European defense landscape in a four-member team and translated the evidence and trade-offs into a focused set of recommendations.

NH66 Fund, P&L Club – Fund Manager

Jan 2025 – Apr 2026

- Managed strategy for an INR 150,000 student fund and mentored three junior analysts; organized two recruitment drives and a training workshop.

Languages

English; Hindi; Tamil; French (basic).